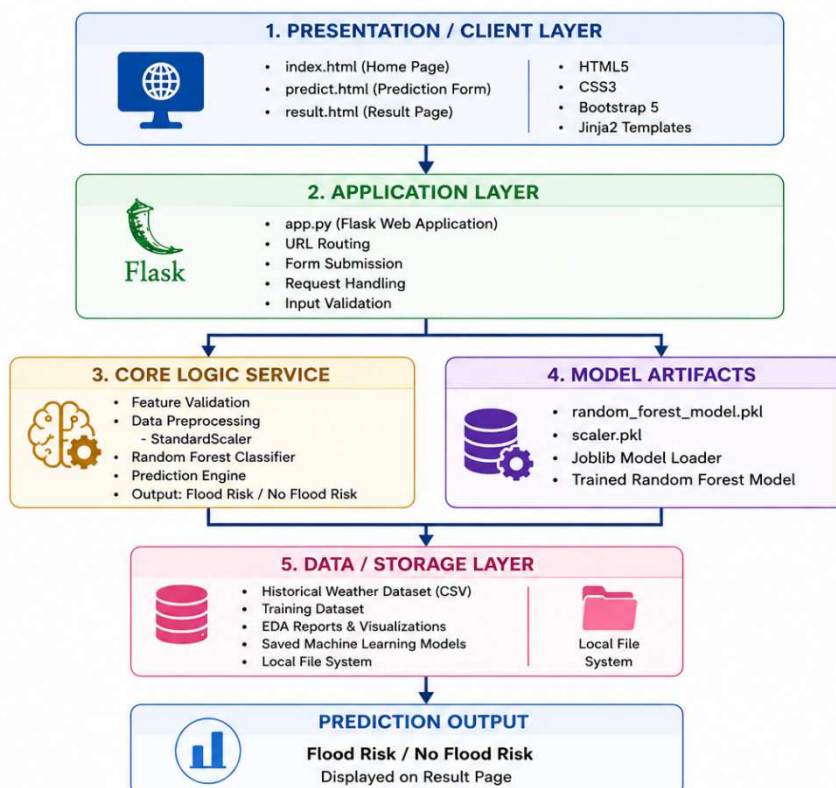


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Rising Water Solution Architecture Diagram:

Solution Architecture – Rising Waters Flood Prediction System



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the interface for entering weather parameters and viewing prediction results.	HTML5, CSS3, Bootstrap, Jinja2
Application Layer	Handles routing, request processing and validation.	Python, Flask
Core Logic Service	Preprocesses data with StandardScaler and predicts using Random Forest.	Scikit-learn, Pandas, NumPy
Model Artifacts	Stores trained model and scaler.	Joblib (.pkl)
Data / Storage Layer	Stores dataset and saved models.	CSV Dataset, Local File System